

Fig. 1: Technical System Architecture — Efficient Temporal Reasoning for News Understanding
Five-stage inference pipeline with offline indexing, temporal-aware RAG, and LoRA-adapted SLMs

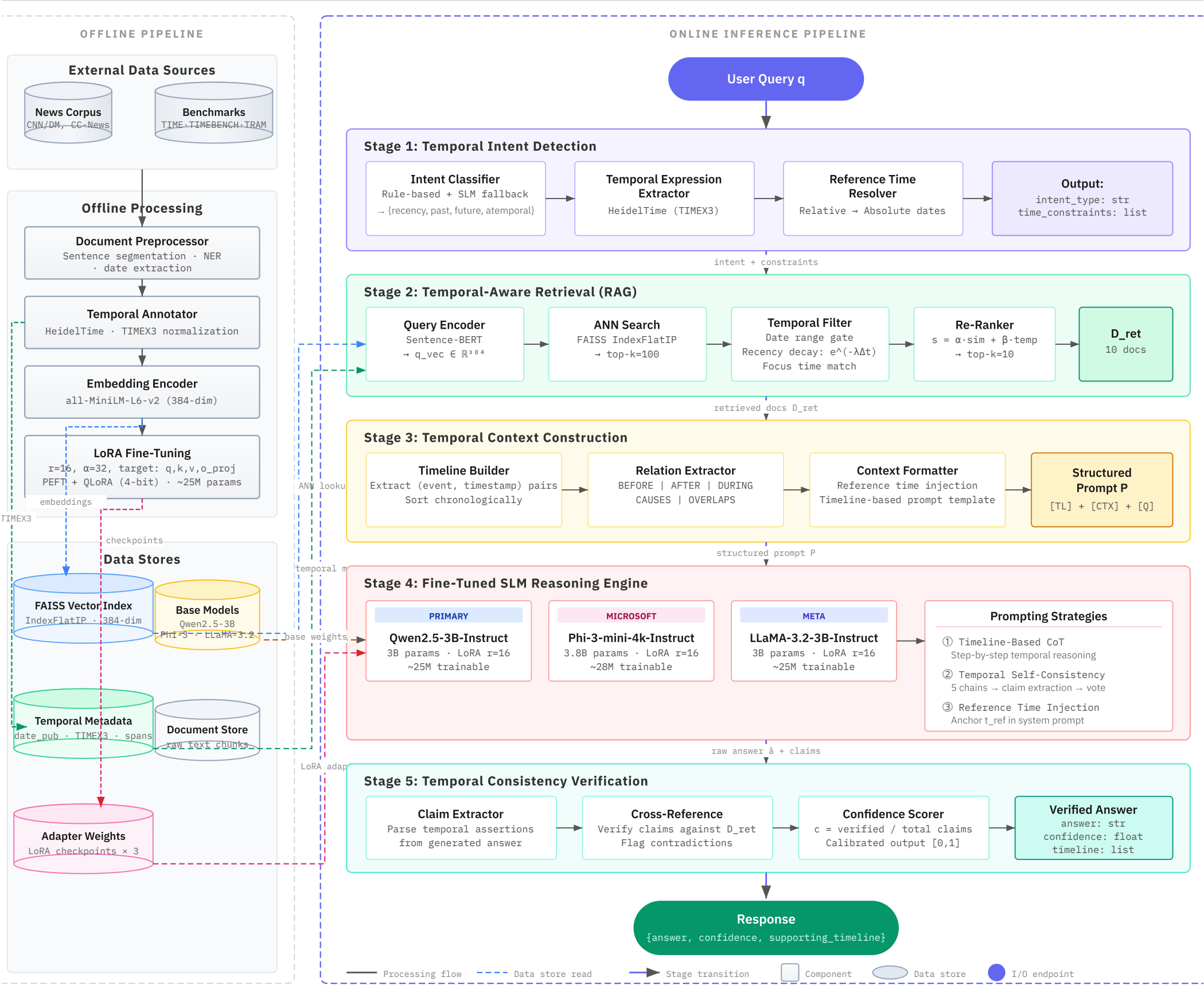


Fig. 1. System architecture of the Efficient Temporal Reasoning pipeline. The *offline pipeline* (left) pre-processes documents, builds the FAISS index and temporal metadata store, and trains LoRA adapters on three SLMs using curriculum-based fine-tuning. The *online inference pipeline* (right) processes a user query through five sequential stages: (1) temporal intent detection, (2) temporal-aware RAG retrieval, (3) context construction with timeline prompt, (4) SLM reasoning with Chain-of-Thought, and (5) consistency verification. All components use open-source tools.